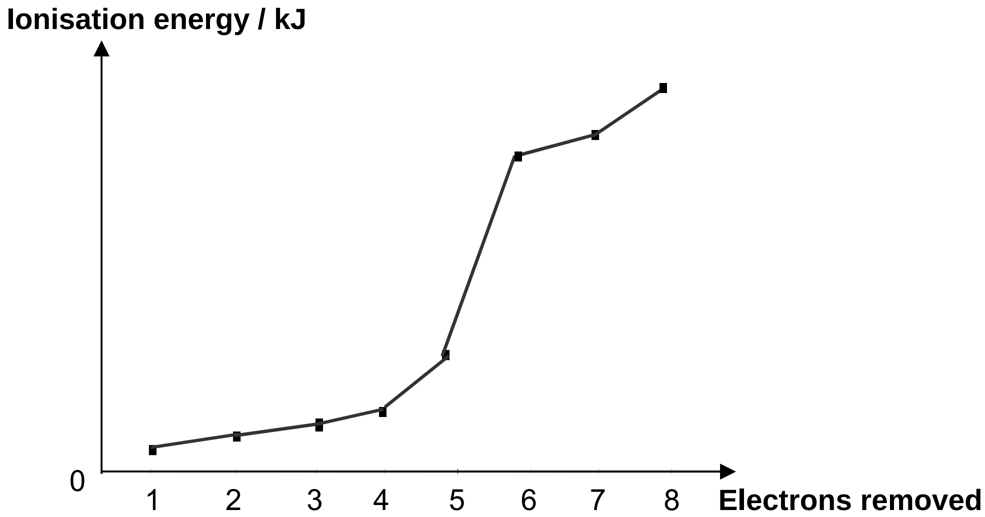
## Section A (40 marks)

Answer **all** the questions in this section in the space provided.

**1 (a)** The first eight ionisation energies of element **X** are shown in the graph below.




**(i)** To which Group does element **X** belong to? Explain your answer.

**X**: Group V

There is a big jump from the 5th IE to the 6th IE, suggesting that there are 5 valence electron and the 6th electron to be removed lies in the inner shell.

**W**, **X** and **Y** are consecutive elements of increasing proton number in Periods 3. These elements forms ions which are *isoelectronic* with element **Ar**.

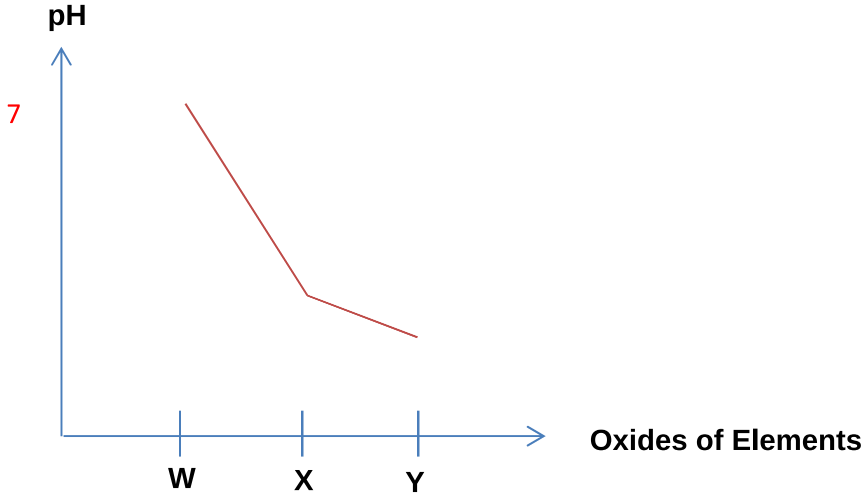
**(ii)** Explain the term 'isoelectronic'.

Isoelectronic refers to species having the same number of electrons.

**(iii)** Write the formula of the ions of **W**, **X** and **Y** which are isoelectronic with element **Ar**.

**W**: $W^{4-}$/ $Si^{4-}$ **X**: $X^{3-}$/ $P^{3-}$ **Y**: $Y^{2-}$ / $S^{2-}$ **[5]**

**1 (b) (i)** Sketch the pH trend of the resultant solutions when the oxides of elements **W**, **X** and **Y** are added to water in the axis below.




**(ii)** Account for the trend above.

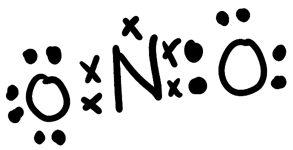
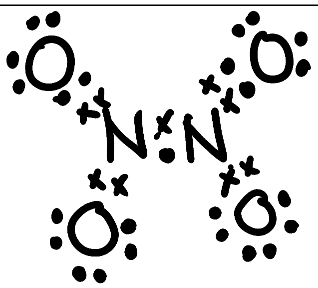
(Oxide of **W**)$SiO_2$ has a giant molecular structure and is insoluble in water. Hence, the pH of the solution is neutral at 7. (Oxides of **X**)$P_4O_{10}$/$P_4O_6$ and (Oxides of **Y**) $SO_2$/$SO_3$ are acidic oxides which will dissolve in water to give acidic solutions. Hence their pH is at 2 - 3. **[4]**

**[Total: 9]**

**2** Nitrogen dioxide, $NO_2$, dimerises to form dinitrogen tetroxide, $N_2O_4$, which is the more stable form in the following equilibrium.

$$2\,NO_2(g) \rightleftharpoons N_2O_4(g) \qquad \Delta H = -57.2\ \text{kJ mol}^{-1}$$

**(a) (i)** Draw the dot-and-cross diagram of $NO_2$ and $N_2O_4$.

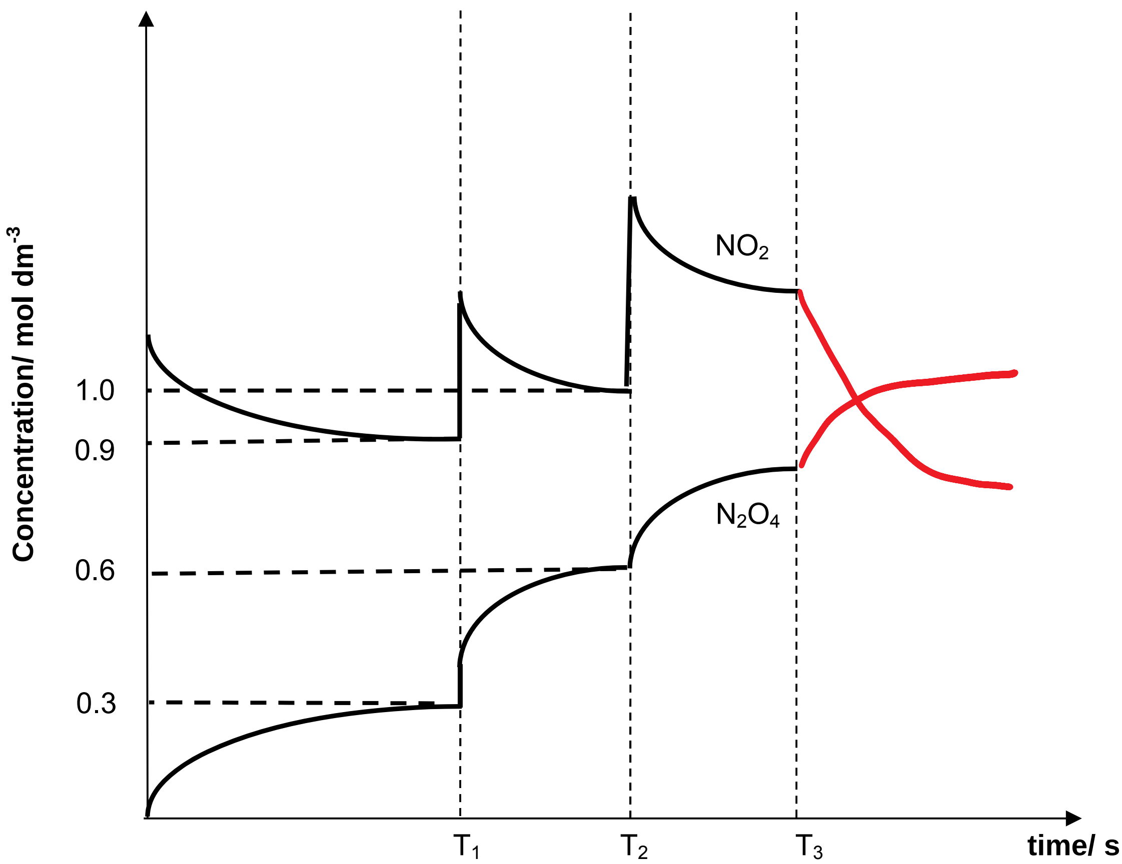
| **$NO_2$** | **$N_2O_4$** |
|---|---|
|  |  |

**2 (a) (ii)** Suggest a reason why the enthalpy change for the forward reaction is exothermic.

Enthalpy change is exothermic as an N–N bond is formed (bond formation) which releases energy. **[3]**

**(b)** During an experiment, changes were made to the conditions in the reaction vessel. At each time, there was only one change made to the condition in the reaction vessel.

The change in the concentrations in the equilibrium mixture with time is shown in the graph below.




**(i)** Write the $K_c$ expression for the equilibrium.

$K_c = [N_2O_4]/ [NO_2]^2$

Note: (no curved brackets)

**2 (b) (ii)** Calculate the $K_c$ at $T_1$, stating the units.

$K_c = 0.3/0.9^2 = 0.37\ mol^{-1}dm^3$ (ecf from (i))

**[3]**

**(c) (i)** What changes in condition were made at time $T_1$ and $T_2$?

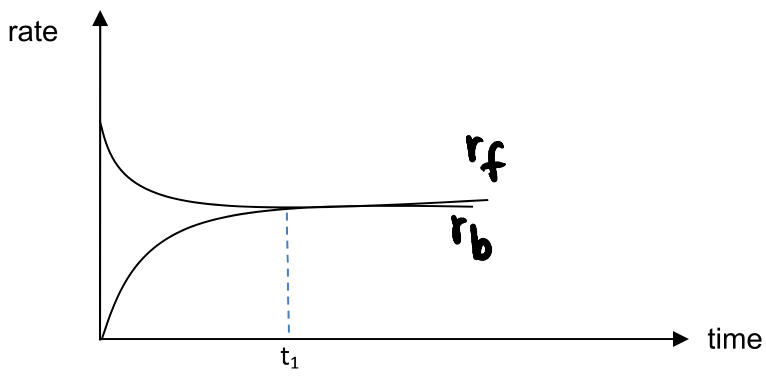
$T_1$: Decrease in volume of vessel/Increase in Pressure

$T_2$: Increase in concentration of $NO_2$

**(ii)** Sketch and label clearly on the given graph on page 4, the changes in the concentration of $NO_2$ and $N_2O_4$, when the mixture was cooled at time $T_3$. Explain your answer.

According to Le Chatelier's Principle, the equilibrium position will shift to the right to favour the exothermic reaction to release heat when temperature is decreased. Hence the mixture will contain more $N_2O_4$ and less $NO_2$. **[5]**

**(d)** Sketch, on the axes below, how the **rates** of the forward and reverse reactions change from the time $NO_2$ was placed in the vessel to the time equilibrium was reached at time $T_1$. Label your graphs clearly.




**[1]**

**[Total: 12]**

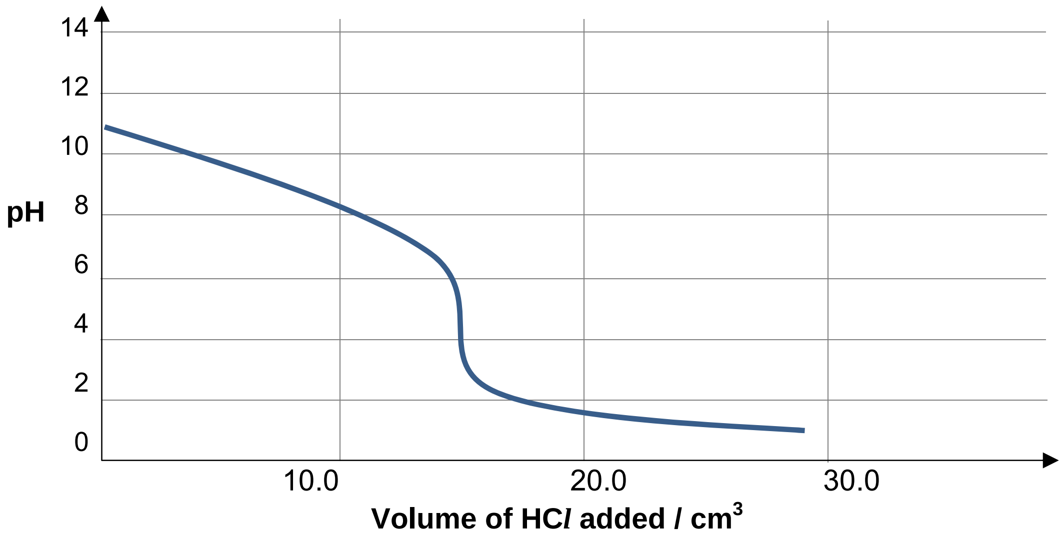
**[Turn Over**

**3** **(a)** Hydrazine, $N_2H_4$, is a flammable liquid with an ammonia like odour. It has basic chemical properties like ammonia.

**(i)** Suggest the equation for the reaction between hydrazine and water, assuming $N_2H_4$ is a monoprotic base.

$N_2H_4 + H_2O \rightleftharpoons N_2H_5^+ + OH^-$

The graph below shows the pH changes when 10.0 $cm^3$ of hydrazine was titrated with 0.05 mol $dm^{-3}$ hydrochloric acid.




| Indicator | p*K*a | acid form | base form |
|---|---|---|---|
| Congo red | 3 - 5 | blue | red |
| Phenol Red | 6.8 - 8.4 | yellow | red |
| Thymolphthalein | 9.3-10.5 | colourless | blue |

**(ii)** From the table above, suggest a suitable indicator for the above titration.

Indicator: Congo Red

**(iii)** Suggest the colour change observed at the end-point.

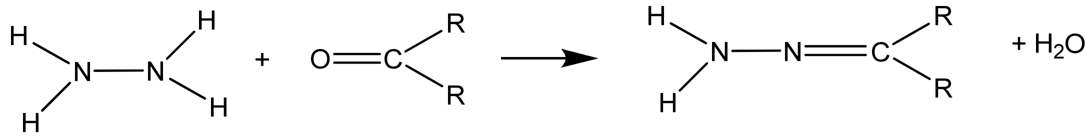
Red to Blue

**[Turn Over**

**3** **(a)** **(iv)** When 7.5 $cm^3$ of HC*l* is added to excess hydrazine, the resultant solution act as a buffer. Write an equation to explain what happens when some solid sodium hydroxide is added to this buffer solution.

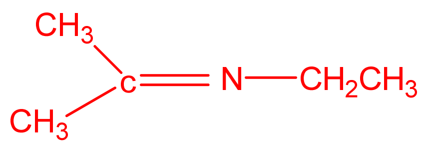
$OH^- + N_2H_5^+ \rightarrow N_2H_4 + H_2O$ **[4]**

**(b)** Hydrazine and its derivatives can undergo condensation reactions with carbonyl compounds as shown.



A similar condensation reaction takes place between $CH_3CH_2NH_2$ and propanone, $CH_3COCH_3$.

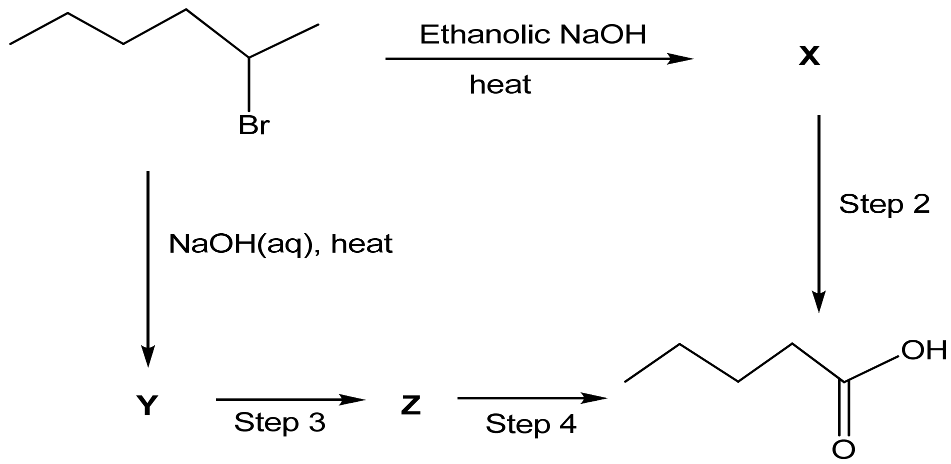
Draw the structural formula of the product formed for the above reaction.



**[1]**

**[Total: 5]**

**4** **(a)** Pentanoic acid could be synthesized from 2-bromo-hexane via two different synthetic routes as shown below, where **X**, **Y** and **Z** are different compounds.



**4 (a) (i)** Suggest the reagents and conditions for **steps 2 – 4**.

**Step 2**: $KMnO_4$(aq), $H_2SO_4$(aq), heat

**Step 3**: NaOH(aq), $I_2$(aq), warm

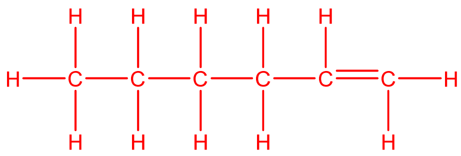
**Step 4**: $H_2SO_4$(aq)

**(ii)** State the type of reaction undergone in **step 3**.

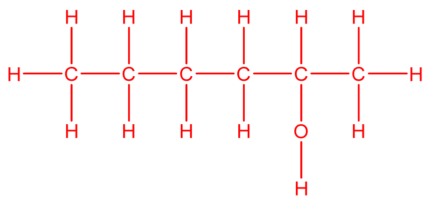
**Type of reaction**: Oxidation

**(iii)** Draw the full structural formulae of Compounds **X, Y** and **Z**.

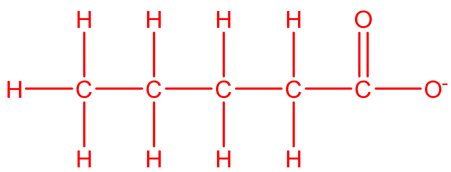
**X**



**Y**



**Z**



**4** **(a)** **(iv)** Suggest a simple chemical test which you will carry out to distinguish between **X** and **Y**.

**Reagents and Conditions:** ______________________________________

**Observations:** ______________________________________

Reagents and Conditions: $PCl_5$ (s), rtp
Observations: white fumes for Y, no white fumes for X.
OR **[9]**
Reagents and Conditions: $I_2$(aq), NaOH(aq), warm
Observations: Yellow ppt for Y, no yellow ppt for X
OR
Reagents and Conditions: $KMnO_4$, NaOH(aq)
Observations: Purple $KMnO_4$ decolourised with formation of brown ppt for X, purple $KMnO_4$ remained purple for Y.
OR
Reagents and Conditions: $K_2Cr_2O_7$, $H_2SO_4$(aq), heat
Observations: Orange $K_2Cr_2O_7$ turned green for Y, orange $K_2Cr_2O_7$ remained orange for X.
OR
Reagents and Conditions: $Br_2$ (aq)
Observations: Orange $Br_2$ decolurised for X, orange $Br_2$ remained orange for Y.
Reagents and Conditions: $Br_2$ in $CCl_4$ in the dark
Observations: Reddish brown $Br_2$ decolurised for X, reddish brown $Br_2$ remained reddish brown for Y.
OR
Reagents and Conditions: $Br_2$ (aq)
Observations: Orange $Br_2$ decolurised for X, orange $Br_2$ remained orange for Y.
OR

**(b)** The following table shows the $pK_a$ values of pentanoic acid, 3-hydroxypentanoic acid and pentanol.

| **Compound** | **$pK_a$** |
|---|---|
| pentanoic acid | 4.87 |
| 3-hydroxypentanoic acid | 3.89 |
| pentanol | 16 |

**(i)** Suggest a reason why the $pK_a$ value of pentanoic acid is less than that of pentanol.

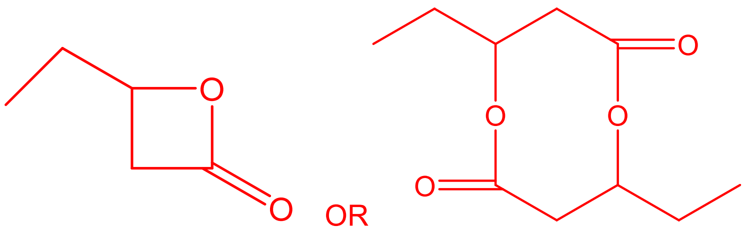
Pentanoic acid is more acidic as the carboxylate ion can disperse the negative charge over the two oxygen atoms and stabilises the anion. For pentanol, the electron donating R group on pentoxide ion intensifies the negative charge and destabilises the anion, making it much less acidic.

**(ii)** Suggest a reason why 3-hydroxypentanoic acid is more acidic than pentanoic acid.

3–hydroxypentanoic acid can form intra–molecular hydrogen bonds between the –OH and the $-COO^-$ and this stabilises the anion.
OR
the electron withdrawing –OH group disperses the negative charge on the anion to a greater extent, hence stabilizing the anion to a greater extent.

**[4]**

**(c)** When concentrated sulfuric acid was added to 3-hydroxypentanoic acid, the mixture was refluxed. Only one organic product was obtained.
Draw the structural formula of the product formed.




**[1]**

**[Turn Over**

## Section B

Answer **two** questions from this section on separate answer paper.

**5** Ethylamine, $CH_3CH_2NH_2$, is a colourless gas with a strong odour. It can be synthesised by reacting ethene with ammonia, as shown in the equation below:

$$CH_2{=}CH_2\,(g) + NH_3\,(g) \longrightarrow CH_3CH_2NH_2\,(g)$$

**(a)** **(i)** State the hybridisation of the C atoms in ethene and in ethylamine.

ethene – $sp^2$

ethylamine – $sp^3$

**(ii)** The C-H bonds in ethene are shorter than the C-H bonds in ethylamine. With reference to the hybridisation theory, explain the difference in bond length. **[4]**

$sp^3$ hybrid orbital has more p character / $sp^2$ hybrid orbital has more s character. Hence the $sp^3$-s overlap is less effective / $sp^2$-s overlap is more effective.

**(b)** Using relevant data from the *Data Booklet*, calculate a value for the enthalpy change for above reaction. **[3]**

Energy required to break bonds = 610 + 4(410) + 3(390)

= 3420 kJ $mol^{-1}$

Energy evolved from forming bonds = 350 + 5(410) + 305 + 2(390)

= 3485 kJ $mol^{-1}$

$\Delta H$ = 3420 – 3485 = -65.0 kJ $mol^{-1}$

**(c)** An experiment was carried out to study the kinetics of the synthesis of ethylamine using ethene and ammonia. The rate equation was found to be

$$\text{rate} = k[CH_2{=}CH_2][NH_3].$$

When $CH_2{=}CH_2$ and $NH_3$, both at an initial concentration of $1.00 \times 10^{-4}$ mol dm$^{-3}$, were mixed together at 20°C, $[CH_3CH_2NH_2]$ varies with time, as shown in the table below.

| **Time / min** | **$[CH_3CH_2NH_2]$ / mol dm$^{-3}$** | **$[NH_3]$ / mol dm$^{-3}$** | **$[NH_3]^2$ / mol$^2$ dm$^{-6}$** | **$[CH_2{=}CH_2]$ / mol dm$^{-3}$** | **Rate / mol dm$^{-3}$ min$^{-1}$** |
|---|---|---|---|---|---|
| 0 | 0 | $1.00 \text{ X } 10^{-4}$ | $1.00 \text{ X } 10^{-8}$ | $1.00 \text{ X } 10^{-4}$ | 0.00 |
| 5 | $5.00 \times 10^{-5}$ | $5.00 \times 10^{-5}$ | $2.50 \times 10^{-9}$ | $5.00 \times 10^{-5}$ | $1.00\times 10^{-5}$ |
| 10 | $6.90 \times 10^{-5}$ | $3.10 \times 10^{-5}$ | $9.61 \times 10^{-10}$ | $3.10 \times 10^{-5}$ | $3.80\times 10^{-6}$ |
| 15 | $7.70 \times 10^{-5}$ | $2.30 \times 10^{-5}$ | $5.29 \times 10^{-10}$ | $2.30 \times 10^{-5}$ | $1.60\times 10^{-6}$ |
| 20 | $8.20 \times 10^{-5}$ | $1.80 \times 10^{-5}$ | $3.24 \times 10^{-10}$ | $1.80 \times 10^{-5}$ | $1.00\times 10^{-6}$ |

**(i)** Explain the meaning of the term *rate equation*.

A rate equation is determined experimentally and it shows the relationship between the rate of reaction and the concentrations of each reactant raised to a specific power.

**(ii)** The rate at a particular time can be approximated using

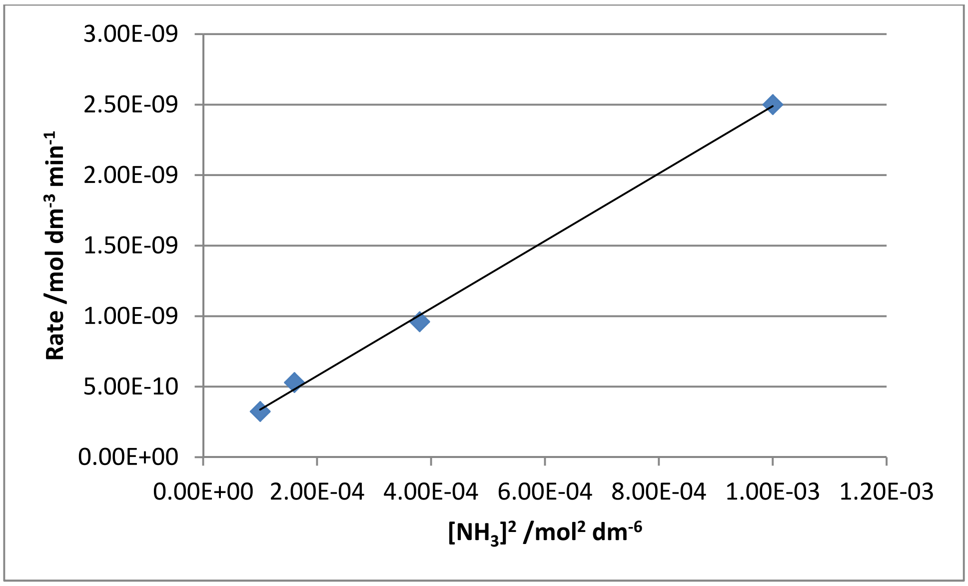
$$\text{rate} = \frac{\Delta[CH_3CH_2NH_2]}{\Delta t}.$$

Reproduce and complete the above table on your writing paper and hence plot a suitable **straight line graph** to confirm that the reaction shows overall second order kinetics.

Since the reaction is overall second order, the following rate equations may be possible:

Rate = k $[NH_3]^2$ or Rate = k $[NH_3][CH_2{=}CH_2]$

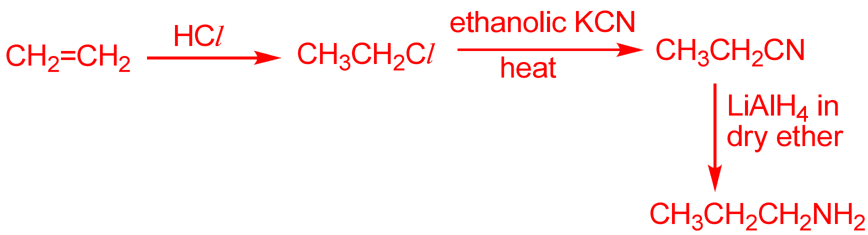
As such, a graph of ***rate against $[NH_3]^2$*** or a graph of ***rate against $[NH_3][CH_2{=}CH_2]$*** would give a straight line. Values for $[NH_3]^2$ and $[NH_3][CH_2{=}CH_2]$ should be identical since $[NH_3] = [CH_2{=}CH_2]$ at every timing.




**5 (c) (iii)** Hence, determine the rate constant *k*, with units. **[8]**

k = gradient of graph = $2.00 \times 10^{-6}$ $mol^{-1}$ $dm^{3}$ $min^{-1}$

**(d)** The yield of propylamine from the reaction between propane and ammonia is low. Propylamine can be produced from ethene via another process. Suggest how you can make propylamine from ethene in 3 steps, showing the reagents and conditions necessary as well as all intermediates. **[5]**




**[Total: 20]**

**6 (a)** A solid sample contains a mixture of sodium oxide and barium oxide only. The percentage by mass of barium oxide in the sample was determined using a titrimetric method.

The sample of unknown mass was reacted with 50 $cm^3$ of 0.0100 mol $dm^{-3}$ of dilute sulfuric acid, forming a white precipitate, barium sulfate, and a colourless solution. The resultant mixture was filtered. Upon drying, the residue weighed 0.0352 g.

The filtrate was titrated against 0.0200 mol $dm^{-3}$ solution of NaOH. 17.30 $cm^3$ of NaOH was required to completely neutralise the excess sulfuric acid.

**(i)** Write two separate equations, with state symbols, for the reactions between the two oxides and sulfuric acid.

$Na_2O$ (s) + $H_2SO_4$ (aq) → $Na_2SO_4$ (aq) + $H_2O$ (*l*)

BaO (s) + $H_2SO_4$ (aq) → $BaSO_4$ (s) + $H_2O$ (*l*)

**(ii)** Calculate the mass of barium oxide present in the sample.

amount of BaO = amount of $BaSO_4$ = 0.0352/233.1 = $1.510 \times 10^{-4}$ mol

mass of BaO = $1.510 \times 10^{-4} \times 153$ = 0.0231 g

**(iii)** Calculate the amount of sulfuric acid that reacted with the sample of solid oxides.

amount of excess $H_2SO_4$ = $0.5 \times 0.0173 \times 0.0200 = 1.73 \times 10^{-4}$ mol

amount of $H_2SO_4$ reacted with the oxides = $0.05 \times 0.01 - 1.73 \times 10^{-4}$

= $3.27 \times 10^{-4}$ mol

**(iv)** Hence calculate the amount of sulfuric acid reacted with solid sodium oxide only.

amount of $H_2SO_4$ reacted with $Na_2O$ = $3.27 \times 10^{-4} - 1.510 \times 10^{-4}$

= $1.76 \times 10^{-4}$ mol

**(v)** Calculate the total mass of the sample of solid oxides and hence the percentage by mass of barium oxide in the sample. **[9]**

mass of $Na_2O$ in the sample = $1.76 \times 10^{-4} \times 62$ = 0.0109 g

total mass of the sample = 0.0109 + 0.0231 = 0.0340 g

percentage by mass of BaO = 0.0231/0.0340 × 100% = 67.9 %

**(b)** Solution **P** was made by dissolving 0.500 g of $Na_2O$ and 0.232 g of BaO in 500 $cm^3$ of water. Solution **P** was then used to determine the concentration of a solution of ethanoic acid via titration.

**(i)** Calculate the pH of solution **P**.

$Na_2O + H_2O \rightarrow 2\ NaOH$

amount of $OH^-$ from $Na_2O$ = 0.500/62 × 2 = 0.01613 mol

$BaO + H_2O \rightarrow Ba(OH)_2$

amount of $OH^-$ from BaO = 0.232/153 × 2 = 0.003033 mol

total amount of $OH^-$ = 0.01916 mol

$[OH^-]$ = 0.01916/0.5 = 0.03833 mol $dm^{-3}$

pOH = -lg $[OH^-]$ = 1.417

pH = 14 – pOH = 12.6

**(ii)** Suggest a suitable indicator for the titration. **[4]**

Phenolphthalein

**(c)** Alcohol **F**, $C_6H_{14}O$, changes orange acidified potassium dichromate (VI) to green, and gives a yellow precipitate with warm aqueous alkaline iodine. **F** reacts with concentrated sulfuric acid at 180°C to give **G** as the only product.

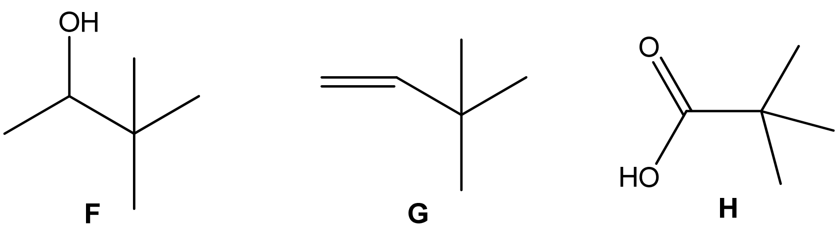
On treating **G** with an excess of hot acidified potassium manganate (VII), compound **H**, $C_5H_{10}O_2$, is formed.

**H** gives effervescence with solid sodium hydrogencarbonate.

Use the information above to deduce the structural formulae for **F**, **G** and **H**, explaining your reasoning. **[7]**

| **Information** | **Deductions** |
|---|---|
| Alcohol **F** changes orange $K_2Cr_2O_7$ to green. | **F** undergoes oxidation. [1/2]<br>**F** contains 1° or 2° alcohol.[1/2] |
| **F** gave yellow precipitate with warm aqueous alkaline iodine. | **F** has the structure $CH_3CH(OH)$-. [1/2]<br>Oxidation. [1/2] |
| **F** reacts with concentrated sulfuric acid at 180°C to give **G** as the only product. | Elimination [½]<br>**G** is an alkene [1/2]<br>Only one neighbouring C of the C atom with the –OH group on **F** has H atom attached to it. [1/2] |
| On treating **G** with an excess of | Oxidation |

| | |
|---|---|
| hot acidified $KMnO_4$, compound **H**, $C_5H_{10}O_2$, is formed. | **H** has one less C atom, C=C of **G** on terminal carbon. [1] |
| **H** gives effervescence with solid sodium hydrogencarbonate. | **H** is a carboxylic acid. [1/2] acid-carbonate / neutralisation reaction [1/2] |




**[Total: 20]**

**7 (a)** Grignard reagents are typically used as a reagent to form alcohols from carbonyl compounds.

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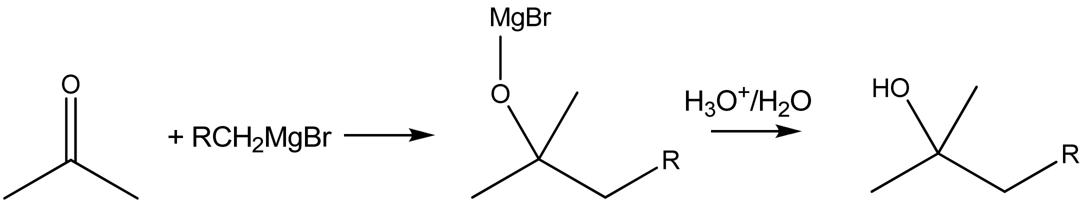



**(i)** Suggest the starting carbonyl compound needed for the formation of pentan-3-ol using $CH_3CH_2MgBr$ as the Grignard reagent.

Propanal

**(ii)** One of the problems associated with the use of magnesium ribbon to produce Grignard reagents is the slow initial step. This is especially so when the magnesium ribbon exposed to air is used. Suggest a reason.

MgO will be formed.

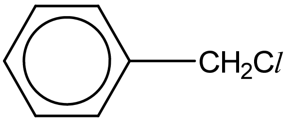
**(iii)** A few drops of Universal Indicator turns yellow in an aqueous solution of $MgBr_2$. Explain with the aid of relevant equations. **[4]**

The high polarising power of $Mg^{2+}$ causes $MgBr_2$ to undergo partial hydrolysis and $H^+$ ions are formed, accounting for the mild acidic solution.

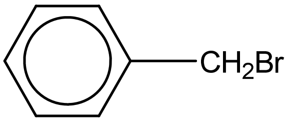
$$MgBr_2\,(s) + aq \rightarrow [Mg(H_2O)_6]^{2+}\,(aq) + 2Br^-\,(aq)$$

$$[Mg(H_2O)_6]^{2+}\,(aq) + H_2O \rightleftharpoons [Mg(H_2O)_5OH]^+\,(aq) + H_3O^+\,(aq)$$

**(b)** Benzyl halides are chemical warfare agents used as tear agents. Two such benzyl halides are benzyl chloride and benzyl bromide.



Benzyl chloride



Benzyl bromide

The boiling points of benzyl chloride and benzyl bromide are given below.

| **Compound** | **Boiling point (°C)** |
|---|---|
| benzyl chloride | 179 |
| benzyl bromide | 201 |

**(i)** State the reagent and conditions needed to convert methylbenzene into benzyl chloride. State the type of reaction involved.

limited $Cl_2(g)$, UV light

Substitution/ Free radical substitution

**(ii)** Suggest a simple test-tube reaction you could use to distinguish benzyl chloride from its isomer, 2-chloromethylbenzene. You should state the reagent used and the observation you would expect to make.

aqueous NaOH, heat. Cool, excess dil $HNO_3$, $AgNO_3(aq)$

OR

Heat with ethanolic $AgNO_3$

Benzyl chloride will give a white ppt, 2-chloromethylbenzene will not give white ppt.

**(iii)** Account for the difference in the boiling points between benzyl chloride and benzyl bromide.

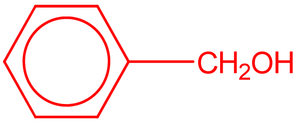
Benzyl bromide has more electrons and hence more polarisable, resulting in a more extensive induced dipole-induced dipole interactions.

More energy required to overcome the stronger id-id in benzyl bromide.

**(iv)** State the reagents and conditions needed to convert benzyl chloride to benzaldehyde, $C_6H_5CHO$, in two steps, drawing any intermediate product formed. **[9]**

Step 1: NaOH (aq), heat

Step 2: $K_2Cr_2O_7$ (aq), $H_2SO_4$ (aq), heat to distill

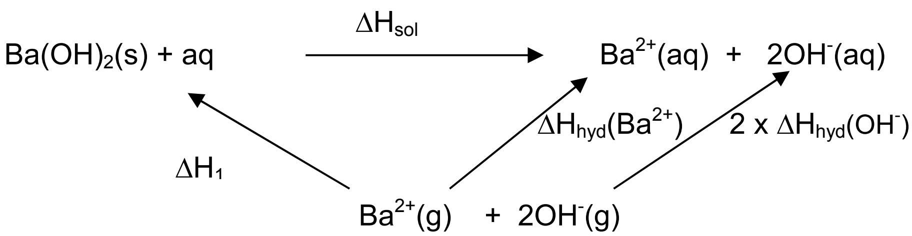
Intermediate: 

**(c)** The enthalpy change of solution of barium hydroxide, $\Delta H_{sol}$, is defined according to the following equation:

$$Ba(OH)_2(s) + aq \longrightarrow Ba^{2+}(aq) + 2OH^-(aq)$$

An energy cycle is constructed using the information in the table below.

| **Enthalpy change** | **kJ mol$^{-1}$** |
|---|---|
| Enthalpy change of hydration of $Ba^{2+}$, $\Delta H_{hyd}(Ba^{2+})$ | −1505 |
| Enthalpy change of hydration of $OH^-$, $\Delta H_{hyd}(OH^-)$ | −664 |
| $\Delta H_1$ | −2850 |




**(i)** Identify $\Delta H_1$.

Lattice energy of $Ba(OH)_2$

**(ii)** Using the information given and the energy cycle above, calculate the enthalpy change of solution for $Ba(OH)_2$. **[3]**

$\Delta H_1 + \Delta H_{sol} = 2\Delta H_{hyd}(OH^-) + \Delta H_{hyd}(Ba^{2+})$

$\Delta H_{sol} = 2 \times (-664) + (-1505) + -2850$

$= +17$ kJ mol$^{-1}$

**7 (d)** The standard enthalpy change of neutralisation was determined experimentally by mixing known volumes of aqueous benzoic acid and aqueous barium hydroxide. The process was known to be only 80% efficient. The following results were obtained.

- Initial temperature = 25.0 °C
- Final temperature = 29.8 °C
- Volume of 1.50 mol $dm^{-3}$ $C_6H_5COOH$ (aq) used = 40 $cm^3$
- Volume of 0.50 mol $dm^{-3}$ $Ba(OH)_2$ (aq) used = 40 $cm^3$

**(i)** Use the above data to calculate the standard enthalpy change of neutralisation of benzoic acid. Assume that the specific heat capacity of all solutions = 4.2 J $K^{-1}$ $cm^{-3}$

$q = mc\Delta T$

= (80)(4.2)(29.8-25)

= 1612.8J

Since $Ba(OH)_2$ is the limiting reagent, no. of moles of $Ba(OH)_2$ = 0.020 mol

$nH_2O$= 0.040 mol

$\Delta H$ = -1612.8 (100/80) / 0.040 = -50400J = -50.4 kJ $mol^{-1}$

**(ii)** Predict, with a reason, if the enthalpy change of neutralisation reaction will be more or less exothermic when hydrochloric acid is used instead. **[4]**

Enthalpy change for neutralisation reaction will be more exothermic.

Hydrochloric acid is a strong acid, hence it dissociates completely and no energy is taken in to dissociate the acid.

**[Total: 20]**